# Lloyd Falltrick

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Software Engineer with experience shipping React/TypeScript features to a distributed platform serving **500,000+ enterprise users**. Built a Python + FFmpeg automation tool that eliminated complex CLI workflows and **cut audio-processing time from hours to minutes** for internal teams. Strong background in accessible 3D web experiences, rendering, and systems-level problem-solving across Unity, Java, and low-level graphics.

## EDUCATION

### University of Brighton — BSC Computer Science

September 2024 - June 2027

## EXPERIENCE

### **Outthink** — Software Engineer Fellow

Jan 2026 - May 2026

- Shipped React/TypeScript features to a distributed security-awareness platform used by **500,000+ enterprise employees**, improving UI consistency and reducing user friction across the portal.
- Worked with **front-line AI technologies** such as **MCP Servers, Github copilot (Context management), Media generation**, and more
- **Collaborated directly** with the Product Manager to integrate token-based controls for colours, images, and text, **reducing duplicated styling logic** and giving way for clients to have more control over components.
- Pair-programmed with a **Principal Engineer** on the system's architecture, state-management patterns with Redux and the following reducers, which assisted greatly with current studies and sped up the Jira tickets relating to feats and bugs as part of onboarding to greatly **improve output**.

## PROJECTS

### Immersive SPA using 3D in web - Javascript/HTML/CSS

- Built a single-page application integrating 3D elements while **preserving DOM structure and accessibility**, with support for 60fps interactions on low-end machines overcoming typical canvas-based limitations using [Three.js]() and CSS3D.
- Delivered a technical talk to the Brighton web community on accessible 3D web design patterns.

### Various Game Prototypes - C#/HLSL

- Created multiple prototypes exploring procedural generation, Gerstner waves, and interactive systems.
- Used rapid prototyping to test new **rendering, vector, and shader techniques** during independent research. Each had quantifiable outcomes, such as sinusoidal layering and render pipelines.

### 3D Raycast Renderer - Java

- Implemented a **raycasting engine** inspired by *Wolfenstein 3D (1992)* to deepen understanding of Java and low-level rendering concepts without using OpenGL or any 3D/game related libraries.

### 2D Game With Procedural Generation - Javascript/HTML/CSS

- Designed and built a fully documented second-year project featuring infinite procedural rooms, shops, item systems, and enemy behaviours.
- Later recreated the concept in Unity C# as a 3D experience.

## TECHNICAL SKILLS

### Languages:

JavaScript(ES6+), TypeScript(TS), Java, Python, C#, SQL, HTML, CSS, HLSL

### Frameworks & Tools:

React, Unity, Git,

### Concepts:

OOP, Agile/Scrum, relational databases, accessibility, 3D rendering, procedural generation, REST APIs, ML/AI

### Other Skills:

3D modelling, game programming

## Hobbies

**Music**

**Drumming**

**Games Programming**

**3D Modelling**